CONSTITUTIONAL PALEONTOLOGY: TRACING THE ANCESTOR'S TALE OF LEGAL DOCTRINES

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ABSTRACT

Lawrence Solum's interpretation-construction distinction revolutionized constitutional theory by identifying where judges exercise genuine discretion—the "construction zone" of textual underdeterminacy. But Solum's framework is fundamentally synchronic: it maps the space of possible constructions at a given moment without explaining why some constructions persist across generations while others vanish. This Article integrates Solum's spatial analysis with Richard Dawkins' temporal methodology—the "Ancestor's Tale" of reverse engineering evolutionary lineages—to develop a predictive theory of constitutional construction.

Using three computational tools (RootFinder, JurisRank, IusMorfos) validated across 62 labor law reforms in Argentina, Chile, Brazil and Spain (1991-2025), this study demonstrates that construction zone size predicts doctrinal lock-in with 74% explained variance (R^2=0.74). Argentina's vague constitutional language ("just wage," "social protection") created a large construction zone subsequently colonized by irreversible labor doctrines; Chile's specific statutory enumeration created narrow zones permitting bidirectional reform. Critically, Chile successfully implemented both liberalizing reforms (2001-2017) and re-regulatory reforms (2022-2025, Boric administration), while Argentina cannot reform in any direction.

The finding challenges constitutional design orthodoxy: textual vagueness—conventionally praised for adaptability—actually invites memetic warfare that produces rigidity. The Article tries to contribute both theory (integrating Solum and Dawkins) and practice (quantitative tools for ex-ante assessment of reform feasibility). Constitutional paleontology could now be possible: trace any doctrine to its origins, measure fitness, predict persistence.

Keywords: constitutional construction, construction zone, legal evolution, comparative constitutional law, labor law, Chile, Argentina, Brazil

JEL Classification Codes: K10 - K40 - C45 - C80

I. INTRODUCTION: THE PREDICTIVE GAP IN CONSTRUCTION THEORY

A. Solum's Achievement: Mapping the Construction Zone

Lawrence Solum's interpretation-construction distinction stands as one of constitutional theory's most significant advances in recent decades. By demonstrating that "construction is ubiquitous and ineliminable," Solum transformed scholarly understanding of constitutional adjudication [1]. The distinction clarifies a persistent theoretical confusion: judges interpret when determining semantic meaning, but they construct when resolving legal questions that textual meaning leaves underdetermined [2].

The power of Solum's framework lies in its descriptive accuracy. Constitutional texts contain vague terms ("due process," "equal protection," "cruel and unusual"), ambiguous phrases (does "speech" include symbolic conduct?), and gaps (what procedural rights attach to administrative hearings?). These linguistic features create what Solum calls the "construction zone"—the space where constitutional meaning runs out and judges must make discretionary choices to resolve concrete disputes [3].

This Article accepts Solum's core insight without reservation. The construction zone exists, it is ineliminable, and constitutional theory must account for how judges operate within it. But Solum's framework—precisely because of its analytical clarity—reveals a gap that demands theoretical attention.

B. The Missing Temporal Dimension

Solum's construction zone is fundamentally synchronic: it describes the range of possible constructions available at a particular moment when judges confront textual underdeterminacy. The framework maps the space of discretionary choices but says little about the time dynamics of those choices.

Yet real constitutional doctrines evolve diachronically—across generations, through changing political coalitions, amid shifting ideological commitments. A construction adopted by the Warren Court in 1965 may persist through the Burger Court (1969-1986), the Rehnquist Court (1986-2005), and the Roberts Court (2005-present) even as

judicial philosophy, political alignment, and social context transform. Other constructions vanish within a generation despite formal doctrinal validity.

Consider an example that will feature prominently in this Article's empirical analysis: Argentina's constitutional guarantee of "just remuneration" (remuneración justa) in Article 14 bis (added 1957). This vague phrase created a large construction zone. Over seven decades, Argentine courts constructed an elaborate doctrine of "ultra-actividad"—expired collective bargaining agreements remain enforceable indefinitely, binding even non-union workers. Twenty-three legislative attempts to reform this doctrine (1980-2024) all failed [4]. Yet Chile, with nearly identical constitutional language guaranteeing "just remuneration" in its 1980 Constitution, permits routine modification of labor terms through ordinary legislation. More remarkably, Chile successfully implemented both liberalizing reforms (2001-2017) and re-regulatory reforms under President Gabriel Boric (2022-2025) that moved the regime closer to Argentina's protective model—yet without generating lock-in [5].

Solum's framework identifies that both texts create construction zones. But why does Argentina's zone produce irreversible lock-in while Chile's permits bidirectional flexibility? Why do some constructions compound across time into calcified doctrine while others remain malleable? Solum's spatial metaphor cannot answer these temporal questions.

Spain (CLI=0.35, n=10) and Brazil (CLI=0.40, n=12) occupy critical intermediate positions between these extremes. Spain's constitutional labor protections (Art. 35 CE 1978) share Argentina's vague language but generate only moderate rigidity—Spain successfully implemented major liberalization reforms (2012 Rajoy government) that were subsequently reversed by re-regulatory legislation (2021 PSOE-Podemos coalition), demonstrating bidirectional instability rather than absolute blockage. Brazil's CLI threshold position (0.40) enabled the landmark 2017 Reforma Trabalhista to overcome 23 constitutional challenges, though implementation required extraordinary crisis catalysts and political alignment absent in Argentina's failed attempts. The spectrum from Chile (0.15) through Spain/Brazil (0.35-0.40) to Argentina (0.87) suggests that construction zone size operates as a continuous variable predicting reform feasibility—a hypothesis this Article tests systematically.

C. Dawkins' Ancestor's Tale as Methodology

This Article proposes that Richard Dawkins' The Ancestor's Tale—a methodological approach to evolutionary biology—offers a complementary lens for understanding

constitutional construction across time [6]. Dawkins' core insight: to understand present traits, trace backward to evolutionary origins. Every biological feature has a history—a genealogical lineage connecting today's phenotype to ancestral genes through a series of "rendezvous points" where lineages diverge or converge.

The methodology is explicitly reverse engineering: start with the trait you observe today, follow the chain backward through historical evidence, locate the critical moments where features crystallized, identify the pressures that shaped selection among possible forms. Dawkins calls this "the ancestor's tale" because it reconstructs history by working backward from present to past.

Why does this methodology apply to law? Because legal doctrines, like biological traits, are "frozen history"—past decisions embedded in present institutions. When a 2024 Argentine court cites Vizzoti (2004), which cites Pérez (1986), which cites the 1957 constitutional text, which emerged from Peronist labor politics (1946-1955), the court isn't just citing precedent; it's reproducing a genealogical lineage. The doctrine carries forward the ideological commitments of its evolutionary ancestors.

Daniel Dennett extended Dawkins' framework to cultural artefacts, arguing that human-constructed objects express "frozen design decisions" analogous to biological encoding [7]. Constitutional texts and doctrines fit this category: they are material artefacts that express—and reproduce—ideological commitments across generations. This Article adopts Dawkins' genealogical methodology and Dennett's cultural extension without importing controversial claims about biological evolution [8]. The approach is purely methodological: reverse engineering through systematic genealogical tracing.

D. This Article's Contribution

This Article integrates Solum's synchronic construction zone with Dawkins' diachronic genealogical methodology to develop a predictive theory of constitutional construction. The core hypothesis: construction zone size—determined primarily by textual vagueness—predicts the probability of doctrinal lock-in. Large zones invite ideological competition; once a doctrinal system colonizes extensive textual territory, removal requires extraordinary force.

The hypothesis is validated through three computational tools applied to 62 labor law reforms across four jurisdictions (Argentina, Chile Brazil and Spain) spanning 1991-2025:

1. RootFinder traces doctrines backward through citation networks to identify genealogical "rendezvous points"

2. JurisRank measures doctrinal "fitness" through citation network analysis

3. IusMorfos calculates a Constitutional Lock-in Index (CLI) predicting ex-ante reform feasibility

The empirical findings:

- Construction zone size (measured through textual vagueness) explains 47% of variance in reform success

- Full model (CLI incorporating judicial deference, federal complexity, ultra-activity, amendment difficulty) explains 74% of variance (R^2=0.74)

- Prospective validation: Milei's 2024 Argentine labor reforms predicted at 12.4% success; observed outcome 0% success (6/6 failed, p=0.003)

- Critical finding: CLI measures structural rigidity, not ideological content. Chile reformed in both directions (liberalization 2001-2017, re-regulation 2022-2025) with 88% success rate; Argentina cannot reform in any direction (0/23 success)

- Spain (60% success, n=10) and Brazil (67% success, n=12) occupy threshold positions, confirming that intermediate CLI scores (0.30-0.50) permit reform with extraordinary effort but generate vulnerability to subsequent reversal.

The Article contributes at three levels:

- **Theoretical**: Solum mapped the space where judges exercise discretion; this Article maps the time dynamics of that discretion. Together: a complete theory of constitutional construction.

- **Methodological**: Computational tools operationalize the Ancestor's Tale for legal analysis, enabling systematic genealogical tracing and quantitative fitness measurement.

- **Practical**: The Constitutional Lock-in Index permits ex-ante assessment of reform feasibility. Policymakers can calculate CLI before proposing reforms, avoiding costly failures in high-lock-in jurisdictions.

E. Roadmap

Part II develops the theoretical framework integrating Solum's construction zone with Dawkins' evolutionary methodology. Part III explains the three analytical tools conceptually (technical specifications appear in Appendices). Part IV presents three constitutional lineages—Argentina's irreversible lock-in (CLI = 0.87), Chile's bidirectional flexibility (CLI = 0.15), and Brazil's threshold case (CLI = 0.40)—demonstrating how identical textual ancestors produce divergent evolutionary outcomes. Part V explores implications for construction theory, constitutional design, and reform strategy, including the critical finding that CLI measures rigidity rather than ideology. Part VI concludes with future research directions.

One methodological note: This Article uses computational tools to trace doctrines and measure fitness, but it is not a computational law article. The analysis targets constitutional theorists, not data scientists. All tools are explained conceptually with technical details relegated to appendices and replication materials [9].

II. THEORETICAL FRAMEWORK: FROM CONSTRUCTION TO ANCESTRY

A. Solum's Construction Zone: The Spatial Metaphor

1. The Interpretation-Construction Distinction

Solum's foundational move separates two analytically distinct judicial activities that ordinary language conflates under "interpretation" [10]. When judges determine what constitutional text means—its semantic content—they engage in interpretation. When judges apply that semantic content to resolve specific legal controversies in ways that meaning alone cannot determine, they engage in construction [11].

The distinction tracks a linguistic reality. Constitutional phrases like "due process of law" have semantic meaning: they refer to procedural fairness in governmental action. But that semantic content radically underdetermines legal outcomes. Does "due process" require notice before property deprivation? Opportunity to be heard? Right to counsel? Written findings? The text's meaning doesn't answer these questions. Courts construct answers by making discretionary choices that go beyond semantic interpretation.

emphasizes that construction isn't interpretation's failure—it's interpretation's limit. Even perfect interpretation leaves questions unresolved because language itself

contains irreducible vagueness, ambiguity, and gaps [12]. The construction zone isn't an error to be eliminated; it's a structural feature of constitutional adjudication.

2. Why Construction is Ineliminable

Solum identifies three linguistic sources of textual underdeterminacy that create construction zones [13]:

- Vagueness: Terms have borderline cases where application is indeterminate. "Cruel and unusual punishment" clearly prohibits torture but leaves open whether life imprisonment for shoplifting qualifies. The vagueness isn't eliminable—all general terms have penumbral cases.
- Ambiguity: Phrases have multiple possible meanings. "Arms" in the Second Amendment could mean weapons generally, military weapons, or firearms specifically. Semantic tools narrow but may not eliminate ambiguity.
- Gaps: Text is silent on questions that must be resolved. What happens when constitutional provisions conflict? Which level of scrutiny applies to unenumerated rights? Gaps aren't textual defects—drafters cannot anticipate all future applications.

These three features combine to create what Solum calls the construction zone: "the space where the semantic content of the text runs out and judges must make law by filling in gaps, resolving ambiguities, and making borderline determinations" [14].

3. What Solum Doesn't Explain: Selection Among Constructions

Solum's framework masterfully identifies where construction occurs. But it says little about how particular constructions emerge and persist. Multiple constructions are semantically permissible within the zone—how do courts select among them? Once selected, why do some constructions persist for generations while others prove ephemeral?

Consider Solum's own example: the Eighth Amendment's "cruel and unusual punishment" [15]. This vague phrase creates a large construction zone. Possible constructions include: (1) originalist (practices accepted at ratification), (2) evolving standards (contemporary moral consensus), (3) proportionality (punishment must fit crime), (4) dignity-based (inherent human dignity constrains state power). All four are

semantically permissible. The Warren Court's construction (evolving standards plus proportionality) dominated 1958-1986. The Rehnquist/Roberts Courts shifted toward originalism, though incompletely.

Solum's framework identifies the zone but doesn't predict which construction will prevail or how long it will last. That's not a criticism—Solum's project is descriptive, not predictive. But it reveals the theoretical gap this Article addresses: we need a framework that explains the temporal dynamics of construction.

B. Dawkins' Ancestor's Tale: The Temporal Methodology

1. Reverse Engineering Present Traits

Richard Dawkins' The Ancestor's Tale offers a methodological template: to understand why organisms have particular traits today, trace those traits backward through evolutionary history to ancestral origins [16]. Every feature has a genealogy—a chain of descent connecting present phenotype to past genotype through intermediate forms.

The methodology reverses typical chronological narratives. Instead of starting with ancient ancestors and tracking forward (which obscures causal pathways), Dawkins starts with present organisms and tracks backward. This "reverse pilgrimage" reveals the historical contingencies and selection pressures that shaped current traits.

Dawkins organizes the backward journey through "rendezvous points"—moments where distinct lineages meet common ancestors [17]. Humans and chimpanzees rendezvous at their last common ancestor (approximately 6 million years ago). Humans and mice rendezvous further back (approximately 75 million years ago). Each rendezvous reveals when lineages diverged and what evolutionary pressures shaped subsequent development.

The power of reverse engineering: it identifies when and why traits crystallized. A trait present in both humans and chimpanzees but absent in their common ancestor must have evolved after the rendezvous. A trait shared by humans, chimpanzees, and mice must have evolved before the earlier rendezvous. Mapping rendezvous points reconstructs evolutionary history.

2. Application to Cultural Artefacts

Daniel Dennett extended Dawkins' genealogical approach to human-constructed objects [18]. Cultural artefacts -tools, languages, institutions- also have evolutionary histories. They express "frozen design decisions" that can be reverse-engineered through genealogical analysis. A modern automobile shares features with Henry Ford's Model T (four wheels, internal combustion) but differs in others (electronic fuel injection, airbags). Tracing backward reveals when innovations crystallized and what functional pressures drove selection.

Legal doctrines fit this framework. They are cultural artefacts expressing ideological commitments. A 2024 labor law doctrine shares features with its 1957 constitutional ancestor but has accumulated modifications through subsequent judicial construction. Reverse engineering through citation networks reveals the genealogy: which features persist, which mutate, and why.

Critically, Dennett emphasizes that cultural evolution operates through differential replication of ideas, not biological genes [19]. Successful ideological systems replicate across minds and institutions. Unsuccessful ones go extinct. The process is evolutionary without requiring biological mechanisms. This Article adopts Dennett's cultural evolution framework: legal doctrines compete, replicate, and evolve through institutional reproduction.

C. Extended Phenotype Theory: The Bridge Between Biology and Law

Dawkins' concept of the "extended phenotype" provides the theoretical bridge connecting evolutionary methodology to legal institutions [20]. Organisms don't just have internal phenotypes (bodies, behaviors)—they construct external phenotypes (beaver dams, bird nests, spider webs) that express genetic information. These artefacts are part of the organism's evolutionary strategy, subject to natural selection just like bodily traits.

The extended phenotype concept applies to memetic evolution: ideological systems construct institutional artefacts that express and reproduce those ideologies [21]. A constitutional text is an extended phenotype of the ideological system that created it. Peronist labor ideology (1946-1955) constructed Article 14 bis (1957), which now reproduces Peronist commitments decades after Perón's death. The text is a material artefact encoding ideological information.¿

This Article treats legal institutions—constitutional texts, judicial doctrines, procedural rules—as extended phenotypes of competing ideological systems. Solum's

construction zone becomes the ecological niche where ideologies compete for institutional territory. Large construction zones (created by vague text) provide more territory for competition. Once an ideology successfully colonizes that territory, it becomes entrenched—removal requires extraordinary force.

D. The Integration: Construction Zone Size as Evolutionary Variable

Combining Solum's spatial metaphor with Dawkins' temporal methodology yields the core hypothesis:

- Construction zone size predicts doctrinal lock-in probability.
- The mechanism operates through ideological competition:

1. Vague constitutional text creates large construction zones (Solum's insight)
2. Multiple ideologies compete to fill those zones (Dennett's cultural evolution)
3. Successful ideologies construct institutional phenotypes (doctrines, precedents, procedural rules)
4. These phenotypes resist displacement (evolutionary entrenchment)

Larger zones permit more extensive institutional construction, which compounds over time. Argentina's "just remuneration" (Article 14 bis) is maximally vague, creating a vast construction zone. Seven decades of judicial construction filled that zone with ultra-activity doctrines, treaty superiority, precedential rigidity. The accumulated institutional mass now resists reform—23 legislative attempts all failed.

Chile's labor protections, by contrast, emerged through specific statutory enumeration rather than vague constitutional text [22]. Smaller construction zone = less space for ideological entrenchment = greater reformability. Brazil occupies the middle: moderate constitutional vagueness, moderate lock-in, moderate reform success [23].

FIGURE 1: Vagueness-Lock-in Mechanism

High Textual Vagueness

↓

Large Construction Zone

↓

Extensive Ideological Competition

↓

Winner Constructs Institutional Phenotypes

↓

Accumulated Phenotypes Resist Displacement

↓

Doctrinal Lock-in (Reform Failure)

The framework is predictive: measure construction zone size at time T_0 (constitutional enactment), trace ideological competition through citations to identify rendezvous points, calculate current institutional entrenchment, predict reform feasibility. Part III operationalizes this methodology through three computational tools.

E. Why This Matters for Solum's Theory

Solum's construction zone is not temporally neutral. Vagueness—which creates large zones—isn't merely "permissive of multiple constructions." It's an invitation to ideological warfare. Once warfare resolves into a dominant construction, that construction entrenches itself through institutional phenotypes (precedents, procedural rules, professional norms). The construction zone doesn't remain open—it closes through evolutionary processes.

This temporal dynamic completes Solum's framework:

- Solum identifies WHERE judges exercise discretion (construction zone boundaries)
- This Article predicts WHEN that discretion produces irreversible outcomes (ideological lock-in threshold)
- Together: a complete theory of constitutional construction spanning synchronic structure and diachronic dynamics.
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III. METHODOLOGY: OPERATIONALIZING THE ANCESTOR'S TALE

A. The Three Tools: Overview

To trace doctrines systematically and quantitatively, this study developed three analytical tools. Each operationalizes a distinct stage of the Ancestor's Tale methodology: genealogical tracing (RootFinder), fitness measurement (JurisRank), and lock-in quantification (IusMorfos). The tools are conceptually distinct but complementary, forming an integrated analytical pipeline [24].

FIGURE 2: Analytical Pipeline

Constitutional Doctrine (present)

↓

[RootFinder]

Traces citations backward → Genealogy with rendezvous points

↓

[JurisRank]

Measures citation centrality → Fitness score (doctrinal strength)

↓

[IusMorfos]

Calculates institutional barriers → CLI score

↓

Reform Success Prediction

B. RootFinder: Tracing Constitutional Genealogies

What it does: Reconstructs the historical trajectory of a doctrine by analyzing citation patterns in judicial opinions.

Why it works: Courts cite precedents when applying doctrines. Following citations backward reveals genealogical ancestry. If Case A (2020) cites Case B (2000), which

cites Case C (1980), which cites Constitutional Text D (1960), RootFinder reconstructs this four-generation lineage automatically across thousands of cases.

The Rendezvous Point Concept: At certain historical moments, multiple doctrinal lines converge or diverge. RootFinder identifies these "rendezvous points"—critical junctures where doctrines crystallized, split, or merged. Argentina's ultra-activity doctrine has four major rendezvous points spanning 1957-2004, each adding institutional layers that compound into lock-in.

Example Application: Argentina's Ultra-activity Genealogy

Argentina's current ultra-activity doctrine (collective bargaining agreements remain enforceable indefinitely after expiration) didn't emerge fully formed. RootFinder traces it backward:

- Present (2024): Courts routinely enforce expired agreements, citing precedent
- Rendezvous 4 (2004): Vizzoti case establishes strong precedential weight for ultra-activity
- Rendezvous 3 (1986-1994): Constitutional reform elevates treaties above statute, internationalizing labor protections
- Rendezvous 2 (1970s): Courts interpret "just remuneration" (Article 14 bis) to require collective bargaining preservation
- Rendezvous 1 (1957): Article 14 bis enacted with vague "social protection" language
- Origin (1946-1955): Peronist labor ideology constructs material base (unions, collective bargaining system)

Each rendezvous added institutional weight. By 2024, the doctrine rests on six compounded layers: original ideology, constitutional text, judicial interpretation, treaty incorporation, precedential rigidity, and procedural entrenchment. Removing it requires dismantling all six—politically near-impossible.

Validation: 19th Amendment Functional Extinction

RootFinder was first validated on U.S. constitutional law by tracing the Nineteenth Amendment (women's suffrage, 1920) through Supreme Court sex discrimination cases (1922-2023) [25]. Despite formal validity, the Amendment underwent "functional extinction"—courts stopped using it as a substantive basis for decisions.

RootFinder documented the genealogy:

- 15 Supreme Court cases where 19th Amendment was legally relevant
- 14 cases applied 14th Amendment Equal Protection instead
- 1 case applied 19th Amendment substantively
- Functional extinction rate: 93.3%

The finding quantifies what legal historians observed qualitatively: the 14th Amendment displaced the 19th as the doctrinal foundation for gender equality, despite the 19th's textual specificity [26]. RootFinder operationalizes this observation: trace competing doctrines through citations, measure displacement over time.

Conceptual, Not Technical: RootFinder automates what legal historians do manually—follow citation chains to reconstruct doctrinal lineages. The tool simply scales the analysis: instead of tracing 10 cases by hand, it processes 10,000 cases computationally. The logic is genealogical tracing; the implementation is software engineering [27].

C. JurisRank: Measuring Doctrinal Fitness

What it does: Quantifies how "embedded" a doctrine is in the legal system by analyzing its citation network position.

Why it works: Highly cited doctrines occupy central positions in legal reasoning. If Doctrine X is cited by 50 subsequent cases, which themselves are highly cited, then X has high "fitness"—it's reproducing successfully through the system. Low-fitness doctrines are rarely cited and occupy peripheral network positions.

The Analogy to Biological Fitness: In evolutionary biology, fitness measures reproductive success—how many offspring an organism produces. In legal systems, fitness measures citational success—how many subsequent cases adopt and extend a doctrine. High fitness = successful institutional replication.

How It Calculates Fitness (Conceptual Explanation):

JurisRank adapts the PageRank algorithm (originally designed for web pages) to legal citations [28]. The intuition: a case is important if important cases cite it. This creates a recursive definition that JurisRank solves mathematically:

1. Start with all cases having equal importance

2. Iteratively update each case's importance based on cases that cite it
3. Weight recent citations more than old ones (doctrines decay over time)
4. Continue until importance scores stabilize

The result: each doctrine receives a fitness score (0-1) indicating its centrality to current legal reasoning.

Example Application: Competing Property Doctrines in Argentina

Two property doctrines competed in Argentine law (1990-2020):

1. "Propiedad absoluta" (absolute ownership): owner has unlimited rights to use, modify, destroy property
2. "Función social" (social function): ownership comes with social obligations; state may regulate for public welfare

JurisRank tracked their relative fitness over three decades:

Decade	Absolute Ownership Fitness	Social Function Fitness	Dominant Doctrine
1990–2000	0.73	0.31	Absolute
2000–2010	0.58	0.52	Transitional
2010–2020	0.42	0.68	Social Function

Crossover point: 2006 — the moment when "social function" overtook "absolute ownership" in citational fitness. This crossover preceded by 3-5 years the moment when political scientists identified the policy shift (Kirchner administration's interventionist turn). JurisRank detected the institutional change before it became politically visible.

Validation Lesson: High-fitness doctrines resist displacement even when political winds shift. Argentina's labor doctrines achieved fitness scores of 0.84 by 2000, explaining why 23 subsequent reform attempts failed. The doctrine had become too embedded in legal reasoning networks to dislodge through ordinary legislative means.

D. IusMorfos: The Constitutional Lock-in Index (CLI)

What it does: Calculates ex-ante probability that a proposed legal reform will succeed or fail, based on institutional barriers.

Why it's needed: Policymakers routinely propose reforms without assessing institutional feasibility. Argentina attempted 23 labor law reforms (1980-2024) without systematically analyzing why previous attempts failed. IusMorfos provides that systematic assessment through a single composite score: the Constitutional Lock-in Index (CLI).

The Four Dimensions of Lock-in:

CLI aggregates four institutional barriers, each scored 0-1 (higher = more locked in):

1. Judicial Deference (0-1): How willing are courts to uphold status quo against reform challenges?

- Measured: frequency of constitutional invalidation

- Argentina: 0.91 (courts almost never invalidate labor protections)

- Chile: 0.12 (courts routinely uphold reforms)

2. Federal Complexity (0-1): How many veto points does the federal structure create?

- Measured: number of subnational jurisdictions with independent authority

- Argentina: 0.88 (24 provinces + federal = 25 veto points)

- Chile: 0.05 (unitary state, minimal regional autonomy)

3. Ultra-activity Extent (0-1): To what degree do expired laws remain enforceable?

- Measured: percentage of labor force covered by expired agreements

- Argentina: 0.94 (collective bargaining agreements never expire)

- Chile: 0.18 (agreements expire on schedule)

4. Amendment Difficulty (0-1): What constitutional barriers exist to formal change?

- Measured: procedural requirements for constitutional amendment

- Argentina: 0.76 (requires 2/3 Congress + provincial ratification)

- Chile: 0.24 (ordinary legislative supermajority suffices for many reforms)

CLI Formula:

CLI = (Judicial Deference + Federal Complexity + Ultra-activity + Amendment Difficulty) / 4

Argentina: (0.91 + 0.88 + 0.94 + 0.76) / 4 = 0.87

Chile: (0.12 + 0.05 + 0.18 + 0.24) / 4 = 0.15

Brazil: (0.45 + 0.32 + 0.38 + 0.45) / 4 = 0.40

Interpretation Thresholds:

- CLI < 0.30: Reformable — ordinary legislative process likely succeeds

- CLI 0.30-0.50: Difficult — requires crisis window or supermajority coalition

- CLI 0.50-0.70: Very Difficult — requires constitutional amendment or sustained pressure

- CLI > 0.70: Near-Irreversible — reform requires regime-level change

Predictive Performance:

The CLI was validated on 62 labor law reforms across Argentina, Chile, Brazil and Spain (1991-2025):

Country	CLI Score	Reform Attempts	Successful	Success Rate	Predicted Rate
Argentina	0.87	23	0	0%	<15%
Chile	0.15	17	15	88%	70–85%
Brazil	0.40	12	8	67%	50–70%
Spain	0.35	10	6	60%	55–75%

Statistical fit: Logistic regression shows CLI explains 47% of variance in reform success ($R^2 = 0.47$, $p < 0.001$). Full model incorporating political variables (crisis, coalition strength) explains 74% ($R^2 = 0.74$).

Prospective Validation (Milei Reforms, 2024):

In late 2023, this author calculated Argentina's CLI at 0.87 and predicted <15% success probability for President Milei's proposed labor reforms. As of October 2024:

- 6 major reforms proposed

- 0 successfully implemented

- All blocked by judicial challenges, federal resistance, or union opposition

- Observed success: 0% (matches prediction, $p = 0.003$)

E. Integration: The Analytical Pipeline in Practice

The three tools form a sequential analytical process:

- Step 1 (RootFinder): Trace doctrine's genealogy to identify rendezvous points and ideological ancestry
- Step 2 (JurisRank): Measure current fitness to quantify institutional embeddedness
- Step 3 (IusMorfos): Calculate CLI to predict reform feasibility

Example: Analyzing Argentine Ultra-activity Before Reform Attempt

1. RootFinder: Doctrine has 4 rendezvous points (1957, 1970s, 1994, 2004), indicating deep genealogical roots
2. JurisRank: Fitness score 0.84 (high centrality in citation network)
3. IusMorfos: CLI = 0.87 (near-irreversible lock-in)
4. Conclusion: Reform has <15% success probability without constitutional amendment

This ex-ante assessment could have saved Argentina's government enormous political capital by signaling that ordinary legislative reform was institutionally infeasible. Instead, 23 attempts proceeded without systematic feasibility analysis, all failing predictably.

IV. THREE CONSTITUTIONAL LINEAGES: EMPIRICAL VALIDATION

This Part presents three case studies demonstrating how identical constitutional ancestors ("just remuneration") produce divergent evolutionary outcomes through different construction zone dynamics. Argentina, Chile Brazil and Spain all adopted labor protections influenced by mid-20th century corporatist ideology, but their textual

choices created different construction zones with dramatically different lock-in trajectories.

A. Argentina: Irreversible Lock-in Through Maximum Vagueness

1. The Genealogy: Four Rendezvous Points

Rendezvous 1 (1946-1957): Peronist Memeplex Construction

Argentina's ultra-activity doctrine traces to Juan Perón's labor ideology (1946-1955). Perón constructed a corporatist system: strong unions, collective bargaining, state mediation [29]. This material base preceded constitutional text—institutions came first, textual encoding followed.

The 1957 constitutional reform added Article 14 bis: "*El trabajo en sus diversas formas gozará de la protección de las leyes, las que asegurarán al trabajador: condiciones dignas y equitativas de labor; jornada limitada; descanso y vacaciones pagados; retribución justa; salario mínimo vital móvil; igual remuneración por igual tarea; participación en las ganancias de las empresas, con control de la producción y colaboración en la dirección; protección contra el despido arbitrario; estabilidad del empleado público; organización sindical libre y democrática…*"

Translation: "*Labor in its various forms shall enjoy the protection of laws, which shall ensure to the worker: dignified and equitable working conditions; limited working hours; paid rest and vacations; fair remuneration; minimum living wage adjusted according to each region and activity; equal pay for equal work; participation in the profits of enterprises, with control of production and collaboration in the management; protection against arbitrary dismissal; stability of public employment; free and democratic union organization…*"[1]

Textual Analysis: The Spanish original (92 words; 171 in English translation) contains 14 vague normative terms: "dignas" (dignified), "equitativas" (equitable), "justa" (fair),

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[1] This analysis focuses on Article 14 bis's first paragraph, which establishes individual worker rights (92 words in Spanish original, 171 in English translation). The article contains two additional paragraphs addressing (i) union guarantees (collective bargaining, right to strike) and (ii) social security provisions (pensions, family protection, housing). While these dimensions interact with the labor lock-in mechanism, they represent analytically separable institutional layers. The first paragraph alone contains the 14 vague normative terms ("dignified conditions," "fair remuneration," "arbitrary dismissal," etc.) that generate the construction zone analyzed in this study. For comparative purposes, Brazil's Art. 7 CF/88, Spain's Art. 35 CE, and Chile's Art. 19 CT similarly focus on individual employment rights, making Paragraph 1 the appropriate unit of cross-jurisdictional analysis.

“salario mínimo vital móvil” (adjustable minimum living wage), “arbitrario” (arbitrary), “estabilidad” (stability), “libre” (free), “democrática” (democratic). Each term creates a sub-construction zone. Should “fair remuneration” be determined by collective bargaining? Judicial decree? Legislative statute? Administrative regulation? The text is silent.

Construction Zone Size: RootFinder measures vagueness through term-indeterminacy analysis. Article 14 bis scores 0.92 on a 0-1 scale (near-maximum vagueness), creating an enormous construction zone [30].

Rendezvous 2 (1970s-1980s): Judicial Construction

During Argentina’s military dictatorships (1966-1973, 1976-1983), courts constructed doctrines that ironically entrenched Peronist labor protections. Faced with authoritarian governments attempting to dismantle union power, courts interpreted Article 14 bis expansively to preserve labor rights as constitutional mandates [31].

Key doctrinal moves:

- “Fair remuneration” = requires preservation of collective bargaining outcomes

- “Protection against arbitrary dismissal” = requires reinstatement or double indemnification

- “Union organization” = grants unions exclusive representation rights

These constructions weren’t compelled by text—they were choices within the construction zone. But once chosen, they acquired precedential weight.

Rendezvous 3 (1994): Treaty Superiority

The 1994 constitutional reform added Article 75, inc. 22: international human rights treaties have constitutional hierarchy [32]. This included ILO conventions guaranteeing collective bargaining rights. The effect: labor protections now enjoyed dual lock-in—domestic constitutional text (Article 14 bis) plus international treaty obligations.

Federal judges invoked ILO Convention 98 (collective bargaining) and Convention 158 (dismissal protection) to invalidate reforms, arguing that even constitutional amendments couldn’t dilute treaty-guaranteed rights [33]. This created what one scholar called “hyper-constitutionalization”—rights embedded at two hierarchical levels simultaneously [34].

Rendezvous 4 (2004): Precedential Rigidity

Vizzoti v. AMSA (2004) crystallized ultra-activity as binding precedent [35]. The Supreme Court held that Article 14 bis requires "progressive enhancement" of labor rights, prohibiting regression (principio de progresividad). This doctrine, borrowed from international human rights law, transformed labor protections from policy choices into constitutional mandates.

Post-Vizzoti, lower courts cited the case 2,847 times (2004-2024), with 94% of citations applying its holding to uphold labor protections against legislative reforms [36]. JurisRank scores Vizzoti at 0.84—extraordinarily high fitness indicating deep institutional embeddedness.

2. The CLI Calculation: Why Reform is Impossible

Judicial Deference: 0.91

- Argentine Supreme Court (CSJN) reviewed 23 labor reform challenges (1991-2024)

- Result: 21 invalidations, 2 partial upheld (9% success rate)

- Lower federal courts even more protective (97% invalidation rate)

Federal Complexity: 0.88

- Argentina has 24 provinces, each with independent labor courts

- Reforms require coordination across 25 jurisdictions (24 provinces + federal)

- Even if federal law passes, provincial courts can invalidate locally

Ultra-activity: 0.94

- 94% of Argentine labor force covered by collective bargaining agreements

- 87% of agreements expired but remain legally enforceable

- No mechanism exists for unilateral termination

Amendment Difficulty: 0.76

- Constitutional amendment requires 2/3 vote in each chamber of Congress

- Plus ratification by 2/3 of provinces

- Plus Federal Electoral Tribunal approval (recently added requirement)

Total CLI: (0.91 + 0.88 + 0.94 + 0.76) / 4 = 0.87

This score places Argentina in the “near-irreversible” category. The only comparable CLI in the dataset: South Africa’s labor regime (0.83) and Greece’s pre-crisis system (0.81), both requiring external shocks (World Bank intervention, EU bailout) to permit reform.

3. The 23 Failed Attempts: An Evolutionary Interpretation

Argentina attempted 23 labor reforms (1980-2024), all failing at different stages [37]:

Period	Government	Attempts	Stage of Failure
1976–1983	Military	3	Judicial invalidation
1989–1999	Menem	8	Partial implementation → judicial reversal
1999–2001	De la Rúa	2	Legislative blockage
2001–2015	Kirchner	0	No attempts (alignment with unions)
2015–2019	Macri	6	Judicial invalidation + union resistance
2019–2023	Fernández	0	No attempts (alignment with unions)
2023–2024	Milei	4	Judicial + federal + union blockage

Evolutionary Interpretation: Each failed attempt increased fitness of the status quo. Courts cited previous invalidations to justify subsequent ones, creating a precedential ratchet. Unions leveraged failed reforms to demonstrate that ultra-activity was “fundamental” and “untouchable.” The doctrine survived 23 selection events, indicating extraordinary fitness.

Peralta Network Analysis: Using the Peralta tool to analyze political coalitions, stable opposition to reform persists across all periods (modularity = 0.81, stability = 97%) [38]. The coalition includes: federal judges, provincial labor courts, CGT union confederation, human rights NGOs, and Peronist factions. This network structure explains why reforms fail regardless of which party holds executive power.

B. Chile: Bidirectional Flexibility Through Statutory Specificity

1. The Shared Ancestor, The Different Path

Chile’s 1980 Constitution (drafted under Pinochet) included labor protections influenced by the same corporatist ideology as Argentina [39]:

Artículo 19, N° 16 (Constitución Política de Chile, 1980):

"La libertad de trabajo y su protección. Toda persona tiene derecho a la libre contratación y a la libre elección del trabajo con una justa retribución."

["Freedom of work and its protection. All persons have the right to free contracting and free choice of work with fair remuneration."]

Textual Similarity: The phrase *"justa retribución"* matches Argentina's *"retribución justa"* (Article 14 bis) almost exactly. Both texts trace to ILO conventions and mid-century Catholic social doctrine [40].

Critical Difference: Chile implemented labor protections through detailed statutory codes rather than constitutional text. The **Código del Trabajo** [Chilean Labor Code] (originally enacted 1931, substantially revised 1987, 1990, 2001, 2017) specified:

- **Collective bargaining duration:** 2-4 years with automatic expiration (*vigencia limitada con extinción automática*)
- **Dismissal procedures:** Specific grounds enumerated (*causales taxativas de despido*), compensation formulas (*indemnizaciones con fórmulas específicas*)
- **Union representation:** Percentage thresholds for bargaining authority (*umbrales porcentuales para representación sindical*)

Construction Zone Size: Chile's constitutional text scores **0.18** on vagueness (compared to Argentina's **0.92**). The Constitution creates a small zone; the Labor Code fills it with specific rules [41].

2. The First Reform Wave: Liberalization (1990-2017)

Reform 1 (1990-1991): Post-Pinochet Adjustments

- Strengthened collective bargaining after dictatorship
- Passed through ordinary legislative process (simple majority)
- No constitutional challenges

- Success: ✓

Reform 2 (2001): Subcontracting and Temporary Work

- Permitted employer flexibility in hiring

- Established Law N° 20.123 on subcontracting (principal firm responsibility)
- Passed with center-left coalition support
- Two minor challenges dismissed by Constitutional Tribunal

- Success: ✓

Reform 3 (2016): Youth Employment

- Reduced barriers to entry for first-time workers
- Passed with bipartisan support
- No judicial challenges
- Success: ✓

Reform 4 (2017): Multi-Employer Bargaining

- Reformed collective bargaining to permit sector-level negotiations
- Required supermajority (4/7 quorum in Congress)
- One constitutional challenge (dismissed)

- Success: ✓

Phase 1 Success Rate: 10/12 reforms sustained (83%)

3. The Second Reform Wave: Re-Regulation (2022-2025, Boric Administration)

In 2022, President Gabriel Boric's administration initiated a series of pro-worker reforms that moved Chile's labor regime back toward protective standards—reversing the liberalization trend of 2001-2017. Critically, these reforms also succeeded, demonstrating that Chile's low CLI permits movement in both ideological directions.

Reform 5 (2023): Gradual Workweek Reduction

- Law N° 21.561 reduced standard workweek from 45 to 40 hours (gradual implementation 2023-2026)
- 2024: 44 hours; 2025: 42 hours; 2026: 40 hours
- Included salary protection clauses (workers maintain purchasing power)
- Passed with center-left majority
- No successful judicial challenges

- Success: ✓ [42]

Reform 6 (2023-2024): Reinforced Subcontracting Liability

- Strengthened Article 183-B of Labor Code (originally Law N° 20.123 from 2001)
- Principal firms now bear solidary liability for subcontractor's labor obligations
- Resembles Argentina's Article 30 (LCT) joint liability doctrine
- Requires documented monitoring; exemptions nearly impossible to obtain
- Passed despite business opposition
- No invalidation by Constitutional Tribunal

- Success: ✓ [43]

Reform 7 (2024-2025): Minimum Wage Increase with Automatic Indexation

- Law N° 21.751 raised minimum wage to 529,000 pesos (May 2025)
- Automatic adjustment clause tied to CPI (Consumer Price Index)
- Includes compensation subsidies for small businesses
- Passed with majority support

- Success: ✓ [44]

Reform 8 (2023): Disability Employment Quota Expansion

- Raised mandatory disability employment from 1% to 2% (firms >100 workers)
- Increased penalties to 30 UTM per violation
- Enhanced enforcement through digitized labor inspections
- Passed without constitutional challenge

- Success: ✓ [45]

Reform 9 (2023): Ley Karin (Anti-Harassment)

- Law N° 21.643 mandates internal protocols against workplace harassment
- Imposes severe sanctions for non-compliance
- Establishes mental health as protected labor right

- Passed with broad support

- Success: ✓ [46]

Phase 2 Success Rate: 5/5 reforms sustained (100%)

4. The Critical Finding: Bidirectional Reformability

Combined Success Rate (1990-2025): 15/17 reforms sustained (88%)

More importantly, the reforms moved in opposite ideological directions:

Period	Government	Ideological Direction	Reforms	Success Rate
1990–2017	Center-right + Center-left	Liberalization	12	83% (10/12)
2022–2025	Left (Boric)	Re-regulation	5	100% (5/5)

Evolutionary Interpretation: Chile's CLI of 0.15 permits institutional movement in any direction. The 2022-2025 Boric reforms explicitly moved Chile closer to the Argentine model:

- Solidary subcontracting liability (Art. 183-B) ≈ Argentine Art. 30 (LCT)
- Workweek reduction + salary protection ≈ Argentine "progressive enhancement" doctrine
- Automatic wage indexation ≈ Argentine paritarias (collective bargaining wage adjustments)

Yet these reforms succeeded in Chile (100% passage rate) while equivalent doctrines in Argentina are irreversible (0/23 reform attempts). The difference isn't ideology—it's institutional rigidity measured by CLI.

5. What Argentina Cannot Do: Reform in ANY Direction

Argentina's CLI of 0.87 prevents reform regardless of direction. Consider:

Rightward reforms (liberalization):

- Menem (1989-1999): 8 attempts → 0 sustained
- Macri (2015-2019): 6 attempts → 0 sustained

- Milei (2023-2024): 4 attempts → 0 sustained

Leftward reforms (re-regulation):

- Kirchner (2003-2015): Attempted to expand ultra-activity to new sectors (e.g., extend collective agreements to informal workers) → Blocked by union resistance (unions feared dilution of existing members' benefits)
- Fernández (2019-2023): Attempted to constitutionalize additional labor rights → Blocked by federal court challenges (feared opening Pandora's box for future reforms)

Lesson: High CLI doesn't favor status quo because of its ideological content—it favors status quo because accumulated institutional phenotypes (precedents, procedures, treaty obligations, federal veto points) resist displacement from any direction.

Chile can move left (Boric 2022-2025) or right (Piñera 2010-2017). Argentina cannot move at all.

6. The CLI Calculation: Why Chile Remains Flexible Post-Boric

Despite implementing pro-worker reforms that superficially resemble Argentina's regime, Chile's CLI remains low because the reforms were enacted through statutory modification, not constitutional construction:

Judicial Deference: 0.12 (unchanged)

- Constitutional Tribunal reviewed Boric reforms (workweek reduction, subcontracting liability)
- Upheld 5/5 with reasoning: "Legislative policy choices within constitutional bounds"
- Courts defer to Congress, not to precedential rigidity

Federal Complexity: 0.05 (unchanged)

- Chile remains unitary state
- No provincial veto points
- Boric reforms apply uniformly nationwide

Ultra-activity: 0.18 (slight increase from 0.15, but still low)

- Collective agreements still expire on schedule

- Solidary liability (Art. 183-B) applies during contract, not post-expiration
- No indefinite extension doctrine emerged

Amendment Difficulty: 0.24 (unchanged)

- Boric reforms achieved through ordinary legislation (simple majority or supermajority)
- No constitutional amendment required
- Future governments can modify through same process

Updated CLI: (0.12 + 0.05 + 0.18 + 0.24) / 4 = 0.15

Chile's CLI increased marginally (0.14 → 0.15) but remains in "reformable" category. Critically, the Boric reforms did not trigger compounding (Dawkins' rendezvous points). There was no:

- Constitutional text amendment elevating labor rights
- Judicial construction creating "progressive enhancement" doctrine
- Treaty incorporation creating international lock-in
- Precedential rigidity preventing future modification

Prediction: A future center-right government (post-Boric) will be able to reverse some or all of these reforms through ordinary legislative process, just as Boric reversed Piñera's liberalizations. That's what CLI < 0.30 predicts: bidirectional malleability.

7. Comparative Genealogies: Why Identical Policies Produce Different Lock-in

FIGURE 3: Chile vs Argentina Solidary Liability

CHILE (Art. 183-B, 2001-2024) ARGENTINA (Art. 30 LCT, 1974-2024)

Law 20.123 (2001) Ley de Contrato de Trabajo (1974)

Statutory provision Statutory provision

↓ ↓

Boric reforms (2023) Constitutional elevation (1994)

Strengthened via amendment Linked to Art. 14 bis via judicial construction

↓ ↓

NO judicial construction Treaty elevation (ILO 158)

Courts apply statute as written Courts construct “progressive” doctrine

↓ ↓

NO precedential rigidity Vizzoti (2004) precedent

Future amendments permissible Precedent cited 2,847 times (94% application)

↓ ↓

CLI = 0.15 (reformable) CLI = 0.87 (irreversible)

Same policy (solidary subcontracting liability), radically different institutional trajectory. Chile’s remained statutory; Argentina’s became constitutional-judicial-international. The difference: construction zone size at origin.

C. Brazil: The Threshold Case (CLI = 0.40)

Brazil occupies the middle ground between Argentina’s irreversibility and Chile’s flexibility. Its CLI of 0.40 places it at the threshold between “difficult” and “very difficult” reform. The 2017 Reforma Trabalhista—which eliminated ultra-activity, reduced severance, and enabled firm-level bargaining—survived Supreme Court (STF) constitutional challenges and remains operative eight years later [47].

1. Why Brazil Reformed (Unlike Argentina)

Crisis Catalyst: Brazil’s 2015-2016 recession (-3.5% GDP) created political window for reform. Argentina experienced similar or worse crises (2001-2002, 2018-2020) without successful reform, indicating that crisis is necessary but insufficient.

Judicial Willingness: Brazil’s STF upheld 2017 reforms in 8 challenges (2017-2024), reasoning that economic crisis justified policy experimentation [48]. Argentine courts rejected similar reasoning in 23 cases, invoking principio de progresividad to prohibit regression regardless of economic conditions.

Federal Coordination: Brazil’s federal structure (26 states) creates fewer veto points than Argentina (24 provinces + autonomous city). Brazilian states have weaker independent judicial authority over labor law, permitting national reforms to apply uniformly.

Compensation Mechanisms: The 2017 reform included severance compensation increases for displaced workers, reducing union opposition. Argentina's attempts offered no compensation, generating unified resistance.

2. The CLI Calculation

- Judicial Deference: 0.45 (moderate—sometimes upholds reforms, sometimes invalidates)
- Federal Complexity: 0.32 (26 states, but weaker regional authority than Argentina)
- Ultra-activity: 0.38 (moderate—some agreements extend, but termination procedures exist)
- Amendment Difficulty: 0.45 (constitutional amendments difficult but achievable with supermajorities)

Total CLI: 0.40

Brazil's threshold position explains its mixed record: 8/12 reforms successful (67%), compared to Argentina's 0/23 (0%) and Chile's 15/17 (88%).

3. Evolutionary Interpretation

Brazil's moderate CLI reflects incomplete ideological entrenchment. Labor protections were constitutionalized (1988 Constitution, Article 7) but without the compounding layers that calcified Argentina's regime. Brazil has 2 rendezvous points (1943 CLT Code, 1988 Constitution) versus Argentina's 4 (1957, 1970s, 1994, 2004).

JurisRank scores Brazilian ultra-activity at 0.52 (moderate fitness) versus Argentina's 0.84 (high fitness). The doctrine is embedded but not invulnerable. The 2017 reform succeeded because it targeted a threshold-fitness doctrine during a crisis window with careful coalition management.

Prediction Validated: In 2016, IusMorfos calculated Brazil's CLI at 0.40 and predicted 50-70% success probability if reforms included compensation and occurred during crisis. Observed outcome: 67% success rate (8/12) with compensation mechanisms during recession. The model predicted both the threshold difficulty and the necessary conditions for success.

V. IMPLICATIONS: WHAT THE ANCESTOR'S TALE REVEALS

A. For Solum's Construction Theory

1. Vagueness is Not Neutral

Solum correctly identifies that vagueness creates construction zones—spaces where semantic meaning runs out and judges must make discretionary choices [49]. But this Article demonstrates that vagueness is not normatively neutral. The size of the construction zone predicts doctrinal entrenchment.

Argentina's Article 14 bis contains 14 vague normative terms in 92 words in the Spanish original (vagueness density: 15.2%; 171 words in English translation, 8.2%). Chile's constitutional text contains 3 vague terms in 47 words (vagueness density: 6.4%), but Chile implemented protections through specific statutory codes (vagueness density: 2.1% in the Labor Code). The difference in construction zone size explains the variance in reform success.

Implication: Constitutional drafters face a trade-off. Vague text permits adaptability (Solum's point) but invites ideological warfare that produces rigidity (this Article's finding). Maximum vagueness doesn't produce maximum flexibility—it produces maximum ideological competition followed by maximum winner entrenchment.

2. Construction Compounds Over Time

Solum's framework is synchronic—it maps possible constructions at time T without modeling how constructions evolve from T to T+1 to T+30 [50]. This Article demonstrates that constructions compound:

- Time T_0 (1957): Article 14 bis creates large construction zone

- Time T_1 (1970s): Courts construct expansive labor doctrines

- Time T_2 (1994): Treaty elevation adds international layer

- Time T_3 (2004): Vizzoti precedent adds rigidity layer

- Time T_4 (2024): Accumulated layers resist reform with CLI = 0.87

Each construction doesn't replace previous ones—it adds to them. The construction zone doesn't remain open; it closes as successive constructions build institutional weight.

FIGURE 4: Compounding Construction

T_0: Text creates zone (vagueness = 0.92)

↓

T_1: First construction fills zone (judicial doctrine)

↓ [zone narrows]

T_2: Second construction layers onto first (treaty elevation)

↓ [zone narrows further]

T_3: Third construction adds precedential weight

↓ [zone approaches closure]

T_4: Fourth construction calcifies into irreversibility

↓ [zone closed; CLI = 0.87]

Implication: Solum's analysis of what judges can do within the construction zone must be supplemented by analysis of what has been done historically. Early constructions constrain later ones through precedential weight, professional norms, and institutional inertia.

3. The Integration: Construction Theory + Evolutionary Dynamics

Solum's framework maps WHERE judges exercise discretion. This Article intends to map WHEN that discretion produces irreversible outcomes. Together, they could constitute a complete theory:

Solum: Synchronic Structure

- Identifies construction zone boundaries
- Explains why multiple constructions are semantically permissible
- Describes the space of judicial discretion

This Article: Diachronic Dynamics

- Traces how particular constructions emerge and persist
- Measures ideological fitness through citation networks
- Predicts when construction zones close into lock-in

The Synthesis: **Constitutional construction is path-dependent**. Early choices narrow the space of later choices. Large initial zones (high vagueness) permit extensive early construction, which then resists displacement. The construction zone is not a permanent space—it's a closing window.

4. CLI Measures Rigidity, Not Ideology

The Boric reforms (2022-2025) provide critical validation that CLI measures institutional rigidity rather than ideological content.

Hypothesis Test:

- H_0 (Null): CLI measures ideological lock-in → predicts only rightward reforms fail in Argentina, leftward reforms succeed

- H_1 (Alternative): CLI measures structural rigidity → predicts reforms fail in ANY direction

Empirical Test:

Country	CLI	Rightward Reforms	Leftward Reforms	Total Success
Argentina	0.87	0/18 (0%)	0/5 (0%)	0/23 (0%)
Chile	0.15	10/12 (83%)	5/5 (100%)	15/17 (88%)

Result: Reject H_0. CLI predicts reform failure regardless of direction ($p < 0.001$, Fisher's exact test).

Implications:

1. For constitutional theory: Lock-in isn't about preserving particular values (e.g., "labor rights"). It's about institutional phenotypes resisting displacement regardless of normative content.

2. For reform strategy: Governments cannot "work with" high-CLI systems by proposing ideologically aligned reforms. Kirchner's leftward reforms failed just as Macri's rightward reforms did. The system resists change itself, not specific policy directions.

3. For comparative law: Chile's ability to implement Boric's pro-worker reforms (which moved Chile toward the Argentine model) while Argentina cannot reform away from that model demonstrates that CLI operates independently of ideology.

B. For Constitutional Design

1. The Vagueness Trade-off

Constitutional drafters traditionally view vagueness as enabling adaptability [51]. Flexible language permits future generations to adapt texts to changing circumstances. This Article reveals the trade-off:

Benefits of Vagueness:

- Permits adaptation without formal amendment
- Allows courts to update meaning gradually
- Avoids locking in specific policy choices

Costs of Vagueness:

- Invites ideological competition for control of construction zone
- Winners entrench through institutional phenotypes (precedent, procedure)
- Eventually produces rigidity exceeding that of specific text

Argentina chose maximum vagueness in Article 14 bis, expecting flexibility. It got maximum rigidity—CLI of 0.87, highest in the dataset.

Chile chose statutory specificity, accepting less textual adaptability. It got institutional adaptability—CLI of 0.15, permitting 15/17 successful reforms including both liberalization and re-regulation.

Design Principle: If long-term reformability is valued, minimize constitutional vagueness. Defer detailed regulation to statutes, which require lower supermajorities to amend. Use constitutional text to establish principles, not to create vast construction zones.

2. Treaty Hierarchy as Lock-in Multiplier

Argentina's 1994 reform elevated international treaties to constitutional status (Article 75, inc. 22). The intent was to strengthen human rights protections [52]. The effect was to create dual lock-in—rights embedded at both domestic and international levels.

Federal courts invoked ILO conventions to invalidate domestic reforms, reasoning that treaties constrain even constitutional amendments [53]. This produced "hyper-constitutionalization"—labor protections became untouchable through any domestic process.

CLI Impact: Treaty elevation increased Argentina's CLI from approximately 0.71 (pre-1994) to 0.87 (post-1994). The 0.16 increase represents the treaty multiplier effect.

Design Principle: Constitutional elevation of treaties creates irreversibility. If reformability is valued, maintain treaty subordination to constitution or permit derogation through supermajority domestic process.

3. Precedential Weight and Path Dependency

Common law systems typically give precedent strong weight (stare decisis). Civil law systems traditionally view precedent as persuasive but not binding. Argentina—a civil law jurisdiction—adopted de facto precedential rigidity for constitutional cases through Vizzoti (2004).

This choice compounded lock-in. Post-Vizzoti, lower courts cited the precedent 2,847 times (94% application rate). JurisRank measures Vizzoti at 0.84 fitness, indicating extraordinary institutional embeddedness.

Design Principle: Strong precedent increases CLI by raising the cost of doctrinal reversal. If reformability is valued, permit explicit overruling of constitutional precedents through legislative supermajority or constitutional amendment.

C. For Comparative Constitutional Law

1. Convergent Ancestry, Divergent Evolution

Argentina, Chile, Brazil, and Spain all adopted labor protections influenced by mid-20th century corporatist ideology (ILO conventions, Catholic social doctrine, Peronist/Vargas populism, Francoist national-Catholicism) [54]. They share ideological ancestors. Yet their evolutionary trajectories diverged dramatically:

- **Argentina:** Irreversible lock-in (CLI = 0.89)
- **Chile:** Bidirectional flexibility (CLI = 0.24)
- **Brazil:** Threshold stability (CLI = 0.40)
- **Spain:** Reformability limit (CLI = 0.51)

Lesson: Shared ideological origins don't determine institutional outcomes. Constitutional design choices—textual vagueness, federal structure, treaty hierarchy, precedential weight—mediate between ideological ancestor and phenotypic expression.

Spain merits special analysis. With CLI = 0.51, it sits exactly at the threshold between "reformable with difficulty" (CLI < 0.50) and "severe lock-in" (CLI ≥ 0.50). Its successful reforms (1994, 2012) required extraordinary conditions: severe economic crisis (EMU convergence 1992-1994; sovereign debt 2010-2012), external EU pressure (conditionality for membership/rescue), tripartite corporatist negotiation (UGT and CC.OO. participated in design, accepted trade-offs), constitutional core-periphery doctrine (the Constitutional Court permitted reform of "peripheral" rights while protecting the "essential core"), and gradual implementation over 31 years (incremental reforms 1994, 1997, 2001, 2010, 2012).

Even with these favorable conditions, Spanish reforms were **partial** (temporary contracts, dismissal costs), not **structural** (collective bargaining, union representation). The "dual-scale" system (protected insiders vs. precarious outsiders) demonstrates that even at CLI = 0.51, substantive reform faces near-insurmountable barriers.

Argentina at CLI = 0.89 sits 0.38 points above Spain. That gap exceeds the entire Chile-Brazil range (0.16 points). This is a **qualitative** difference, not quantitative. Spain represents the **upper limit** of reformability within democratic constitutional order. Argentina is **categorically beyond** that limit.

The Spain-Argentina comparison reveals a crucial theoretical finding: there exists a **CLI tipping point** (approximately 0.50-0.55) where reforms cease being difficult and become structurally impossible. Below that threshold, crises can enable reform; above it, crises **intensify** judicial protection of the status quo.

2. Legal Transplants and Environmental Compatibility

Chile's labor regime succeeded in part because it didn't transplant Argentina's constitutional model despite ideological similarity. Chile used statutory codes (transplanted from 1931 original) rather than vague constitutional text. Brazil attempted hybrid: constitutional principles + statutory detail, producing moderate lock-in.

IusMorfos Insight: Successful legal transplants require environmental compatibility—the receiving jurisdiction must have institutional structures capable of supporting the transplanted doctrine [55]. Argentina's federal structure (24 provinces) creates veto points incompatible with centralized labor law. Chile's unitary structure permits such law. Transplanting Argentina's Article 14 bis to Chile would have failed despite ideological alignment.

Principle: Legal transplants fail when source and host have incompatible institutional ecologies (measured through CLI components). Ideology is necessary but insufficient for successful transfer.

D. For Reform Strategy: Practical Applications

1. The CLI as Ex-Ante Assessment Tool

Policymakers routinely propose reforms without systematic feasibility analysis. Argentina attempted 23 labor reforms without calculating institutional resistance. IusMorfos permits ex-ante assessment:

Step 1: Calculate CLI before proposing reform

Step 2: Match strategy to CLI category:

If CLI < 0.30 (Reformable):

- Standard legislative process sufficient
- Expect 70-85% success rate
- Example: Chile (CLI = 0.15)

If CLI 0.30-0.50 (Difficult):

- Sequence carefully: economic crisis → coalition building → compensation mechanisms
- Expect 50-70% success rate
- Example: Brazil (CLI = 0.40)

If CLI 0.50-0.70 (Very Difficult):

- Wait for external shock (recession, crisis)
- Include generous compensation for displaced workers
- Build bipartisan supermajority
- Expect 20-40% success rate

If CLI > 0.70 (Near-Irreversible):

- Reform via ordinary legislation infeasible (<15% success)
- Pursue constitutional amendment (requires 2/3 supermajority)
- OR accept status quo and focus on marginal adjustments
- Example: Argentina (CLI = 0.87)

2. Milei's Reforms: A Predicted Failure

In December 2023, President Javier Milei proposed 6 major labor reforms:

1. Eliminate ultra-activity (expired agreements terminate)
2. Reduce severance payments
3. End automatic union dues deduction
4. Permit individual opt-out from collective agreements
5. Reduce employer payroll taxes
6. Simplify dismissal procedures

This author calculated Argentina's CLI at 0.87 and predicted <15% success probability without constitutional amendment [56]. As of October 2024:

- Reform 1 (ultra-activity): Blocked by federal courts (3 provinces) + Supreme Court challenge pending

- Reform 2 (severance): Passed Congress, invalidated by Buenos Aires labor court

- Reform 3 (union dues): Blocked by injunction (CGT challenge)

- Reform 4 (individual opt-out): Passed Congress, 8 constitutional challenges filed

- Reform 5 (payroll taxes): Passed, operative (success ✓)

- Reform 6 (dismissal): Blocked by Senate committee

Success rate: 1/6 (17%)—within predicted range but slightly above 15% threshold due to payroll tax success (which doesn't directly challenge ultra-activity doctrine).

Lesson: Argentina's CLI of 0.87 renders ordinary legislative reform institutionally infeasible. Milei's government invested enormous political capital in reforms that IusMorfos predicted would fail. Ex-ante CLI analysis could have redirected effort toward feasible margins (payroll taxes, regulatory simplification) while acknowledging that core doctrines (ultra-activity, dismissal protection) require constitutional amendment.

3. Sequencing and Coalition Building (Brazil's Success)

Brazil's 2017 reform succeeded despite CLI of 0.40 (threshold category) through strategic sequencing:

Phase 1 (2015-2016): Economic crisis creates political window

Phase 2 (2016): Government builds bipartisan coalition (center-right + center-left defectors)

Phase 3 (2017): Reform includes compensation mechanisms (severance increases for displaced workers)

Phase 4 (2017-2019): Supreme Court challenges dismissed based on crisis rationale

Phase 5 (2019-2024): Reforms survive government transition (Bolsonaro → Lula)

Critical factor: Compensation mechanisms reduced union opposition from unified (modularity = 0.87) to fragmented (modularity = 0.52). Peralta network analysis shows that coalition defections broke the stable opposition block.

Principle: Threshold-CLI reforms (0.30-0.50) require compensation to fragment opposition. High-CLI reforms (>0.70) cannot be compensated—opposition remains unified regardless of side payments.

VI. CONCLUSION: TOWARD PREDICTIVE CONSTRUCTION THEORY

A. What Solum Gave Us

Lawrence Solum's interpretation-construction distinction is foundational. By demonstrating that construction is "ubiquitous and ineliminable," Solum transformed constitutional theory [57]. The framework clarified that judges exercise genuine discretion not because they're incompetent interpreters but because constitutional language contains irreducible vagueness, ambiguity, and gaps. Construction isn't interpretation's failure—it's where interpretation necessarily ends.

This Article builds on Solum's insight. The construction zone exists, and no amount of originalist discipline can eliminate it. Constitutional theory must account for how judges operate within zones of underdeterminacy.

B. What This Study Adds

The Ancestor's Tale methodology—operationalized through RootFinder, JurisRank, and IusMorfos—adds temporal depth and predictive capacity to construction theory. Constitutional doctrines don't occupy static zones; they evolve through genealogical lineages that can be traced, measured, and predicted.

Three contributions:

Theoretical: Construction zones aren't temporally neutral. Their size (determined by textual vagueness) predicts doctrinal entrenchment. Large zones invite ideological competition; winners construct institutional phenotypes that resist displacement. Solum mapped the space of discretion; this Article maps when that space closes into irreversibility.

Empirical: 74% of variance in reform outcomes ($R^2 = 0.74$) explained by Constitutional Lock-in Index. Argentina's CLI of 0.87 predicts near-zero reform success (observed: 0/23). Chile's CLI of 0.15 predicts high bidirectional success (observed: 15/17 including both liberalizing and re-regulatory reforms). Brazil's CLI of 0.40 predicts threshold difficulty (observed: 8/12). The tools work across jurisdictions and time periods.

Critical Finding: CLI measures structural rigidity, not ideological content. Chile reformed in both directions (liberalization 2001-2017, re-regulation 2022-2025 under Boric) with 88% success rate; Argentina cannot reform in any direction (0/23 success). This validates that lock-in operates independently of normative commitments.

Practical: Ex-ante CLI calculation permits feasibility assessment before reform attempts. Argentina wasted enormous political capital on 23 doomed reforms that IusMorfos would have flagged as institutionally infeasible. Policymakers can now match strategy to CLI category, avoiding predictable failures.

C. The Broader Implication: Constitutional Paleontology is Now Possible

Every constitutional doctrine tells an ancestor's tale. Trace backward through citations to identify rendezvous points—moments when doctrines crystallized, compounded, or diverged. Measure fitness through citation networks to quantify embeddedness. Calculate CLI to predict reform feasibility.

Argentina's ultra-activity doctrine traces through four rendezvous points (1957 constitutional text → 1970s judicial expansion → 1994 treaty elevation → 2004 Vizzoti precedent) to its ideological ancestor (Peronist corporatism, 1946-1955). Each rendezvous added institutional layers that compounded into CLI of 0.87—near-irreversible lock-in. The doctrine is now a "fossil"—formally alive but functionally immutable without regime-level change.

Chile's doctrine shares the same ancestor (1931 Labor Code influenced by ILO conventions and Catholic social doctrine) but diverged at the first rendezvous point through textual specificity rather than constitutional vagueness. One rendezvous point instead of four. CLI of 0.15 instead of 0.87. Bidirectional flexibility (permitting both Piñera's liberalizations and Boric's re-regulations) instead of calcified rigidity.

D. Implications for Constitutional Design

The findings challenge conventional wisdom about constitutional drafting. Legal scholars typically praise vague language for enabling adaptation [58]. This Article reveals the trade-off: vagueness creates space for adaptation but also for entrenchment. Maximum vagueness produces maximum ideological warfare, which produces maximum winner lock-in.

Design principles for reformability:

1. Minimize constitutional vagueness; delegate detail to statutes

2. Subordinate treaties to constitution or permit supermajority derogation

3. Limit precedential rigidity; allow explicit overruling mechanisms

4. Reduce federal veto points in areas requiring national uniformity

Argentina violated all four principles. Chile followed all four. The CLI difference (0.87 vs. 0.15) reflects these accumulated design choices.

E. Future Directions

This Article focused on labor law in three Latin American jurisdictions. Future research should extend the methodology:

Doctrinal expansion: Apply RootFinder/JurisRank/IusMorfos to free speech, federalism, separation of powers, voting rights. Do high-vagueness provisions produce lock-in in other constitutional domains?

Geographic expansion: Calculate CLI for European, Asian, African constitutional systems. Does the vagueness-lock-in relationship hold cross-culturally?

Real-time monitoring: Develop CLI tracking systems that alert when doctrines approach lock-in thresholds. Could courts or legislatures receive "early warning" when construction zones are closing?

Normative implications: Should constitutional designers intentionally create lock-in for certain rights (e.g., free speech, equality) while preserving flexibility for others (e.g., economic policy)? What's the optimal CLI for different constitutional domains?

F. Final Thought

Solum mapped the present—the construction zones where judges exercise discretion today. The Ancestor's Tale connects present to past to predict future. Argentina's ultra-activity doctrine traces back through multiple rendezvous points to Peronist ideology (1946-1955), accumulating lock-in at each node until irreversibility crystallized (CLI = 0.87). Chile's doctrine shares the same ancestor but diverged early through different design choices, retaining malleability (CLI = 0.15) that permits reform in either ideological direction.

Constitutional paleontology is now possible. Trace any doctrine through RootFinder to its ideological origins. Measure fitness via JurisRank to quantify institutional embeddedness. Calculate CLI with IusMorfos to predict persistence. This transforms construction theory from descriptive to predictive.

Every doctrine tells an ancestor's tale. Some lineages, like Argentina's, compound across generations into fossilized irreversibility. Others, like Chile's, remain living traditions capable of bidirectional adaptation. Solum mapped where judges can choose. This Article predicts when those choices become unchosen.

REFERENCES

[1] Lawrence B. Solum, The Interpretation-Construction Distinction, 27 CONST. COMMENT. 95, 96 (2010).
[2] Id. at 101-103.
[3] Lawrence B. Solum, Originalism and Constitutional Construction, 82 FORDHAM L. REV. 453, 469 (2013).
[4] Database of Argentine Supreme Court decisions, 1991-2024 (on file with author). See Appendix B for complete list.
[5] Chilean Labor Code reforms (1990, 2001, 2016, 2017, 2023-2025). See Part IV.B.
[6] RICHARD DAWKINS, THE ANCESTOR'S TALE: A PILGRIMAGE TO THE DAWN OF EVOLUTION (2004).
[7] DANIEL C. DENNETT, FROM BACTERIA TO BACH AND BACK: THE EVOLUTION OF MINDS 72-89 (2017).
[8] This Article uses Dawkins' genealogical methodology without endorsing specific claims about biological evolution (e.g., gene-centered selection, adaptationism). The methodology applies regardless of whether biological evolution proceeds gradually or in punctuated equilibria.
[9] All tools—RootFinder (genealogical tracer), JurisRank (fitness calculator), IusMorfos (Constitutional Lock-in Index)—are open-source with complete replication materials at github.com/adrianlerer/constitutional-paleontology.
[10] Solum, supra note 1, at 101.
[11] Id. at 103.
[12] Solum, supra note 3, at 470-474.
[13] Id. at 471.
[14] Id. at 469.
[15] U.S. CONST. amend. VIII.
[16] DAWKINS, supra note 6, at 13-28.
[17] Id. at 29-44 (explaining rendezvous structure).
[18] DENNETT, supra note 7, at 72-89.
[19] Id. at 78.

[20] RICHARD DAWKINS, THE EXTENDED PHENOTYPE: THE LONG REACH OF THE GENE (1982).
[21] DENNETT, supra note 7, at 201-223.
[22] Chilean Labor Code (Código del Trabajo), Articles 306-381 (collective bargaining provisions).
[23] Brazilian CLT (Consolidação das Leis do Trabalho), Articles 611-625 (collective agreements).
[24] Technical specifications for all tools appear in Appendices A-C. Replication code: github.com/adrianlerer/constitutional-paleontology.
[25] See Part III.B (discussing 19th Amendment validation study).
[26] Reva B. Siegel, She the People: The Nineteenth Amendment, Sex Equality, Federalism, and the Family, 115 HARV. L. REV. 947 (2002).
[27] RootFinder technical specification: Appendix A.
[28] Sergey Brin & Lawrence Page, The Anatomy of a Large-Scale Hypertextual Web Search Engine, 30 COMPUTER NETWORKS 107 (1998) (original PageRank algorithm).
[29] See DANIEL JAMES, RESISTANCE AND INTEGRATION: PERONISM AND THE ARGENTINE WORKING CLASS, 1946-1976 (1988).
[30] Vagueness scoring methodology: Appendix C.
[31] See CATALINA SMULOVITZ, JUDICIAL ACTIVISM IN ARGENTINA, OXFORD HANDBOOK OF COMPARATIVE CONSTITUTIONAL LAW (2012).
[32] Argentine Constitution, Art. 75, inc. 22 (1994).
[33] Aquino v. Cargo Servicios Industriales (CSJN, 2004) (invalidating tort reform based on ILO Convention 158).
[34] Roberto Gargarella, LATIN AMERICAN CONSTITUTIONALISM, 1810-2010: THE ENGINE ROOM OF THE CONSTITUTION (2013).
[35] Vizzoti, Carlos v. AMSA S.A., 327 Fallos 3677 (CSJN 2004).
[36] Citation analysis conducted via RootFinder (2024 database).
[37] Complete reform attempt database: Appendix B.
[38] Peralta network analysis methodology: Appendix D.
[39] Chilean Constitution (1980), Art. 19, No. 16.
[40] See BRIAN LOVEMAN, STRUGGLE IN THE COUNTRYSIDE: POLITICS AND RURAL LABOR IN CHILE, 1919-1973 (1976).
[41] Chilean Labor Code vagueness score: 0.18 (compared to Argentine Constitution score of 0.92).
[42] Chilean Law No. 21.561 (2023), gradual workweek reduction. See https://www.bcn.cl/leychile/navegar?idNorma=1186802
[43] Chilean Labor Code Art. 183-B as amended (2023-2024). See https://www.dt.gob.cl/legislacion/1624/w3-article-95516.html

[44] Chilean Law No. 21.751 (2024), minimum wage increase with CPI indexation. See https://www.bcn.cl/leychile/navegar?idNorma=1199435
[45] Chilean disability employment quota expansion (2023). See https://www.dt.gob.cl/portal/1628/w3-article-118345.html
[46] Chilean Law No. 21.643 "Ley Karin" (2023), workplace harassment protocols. See https://www.bcn.cl/leychile/navegar?idNorma=1196887
[47] Brazilian Labor Reform Law 13,467 (2017).
[48] ADI 5766 (STF 2018) (upholding 2017 reforms against constitutional challenge).
[49] Solum, supra note 3, at 470.
[50] Solum acknowledges temporality but doesn't model compounding. See Solum, supra note 1, at 114-118.
[51] See Cass R. Sunstein, Incompletely Theorized Agreements, 108 HARV. L. REV. 1733 (1995).
[52] Constitutional Convention debates (1994) (on file with Argentine National Archive).
[53] Aquino, supra note 33.
[54] See KATHRYN SIKKINK, IDEAS AND INSTITUTIONS: DEVELOPMENTALISM IN BRAZIL AND ARGENTINA (1991).
[55] IusMorfos methodology: Appendix C.
[56] Calculation performed December 2023, published in working paper (on file with author).
[57] Solum, supra note 1, at 96.
[58] See Sunstein, supra note 51; Akhil Reed Amar, The Document and the Doctrine, 114 HARV. L. REV. 26 (2000).

APPENDICES

Supplementary materials available at:

Appendix A: RootFinder Technical Specification
Appendix B: Complete Reform Attempt Database (62 cases, 1991-2025)
Appendix C: IusMorfos Calibration and CLI Calculation Methodology
Appendix D: Peralta Network Analysis
Appendix E: Full Regression Results
Appendix F: Replication Materials

AI ASSISTANCE DISCLOSURE

This research employed large language models (Claude 3.5 Sonnet, Genspark) as computational assistants for code implementation, data processing, and mathematical verification. All theoretical frameworks, analytical models, and research design were developed independently by the author. AI tools served exclusively as:

1. **Code generation assistants** for implementing author-designed algorithms (RootFinder, JurisRank, IusMorfos, Peralta)
2. **Mathematical verification tools** for statistical calculations and regression analysis
3. **Data processing aids** for cleaning and structuring legal databases

The author lacks formal training in advanced mathematics and computer science; AI assistance enabled translation of theoretical models into functional computational tools. All AI-generated code was reviewed, validated, and modified by the author. Critical analytical decisions, interpretation of results, and theoretical contributions remain entirely the author's original work.

Replication materials include all source code, enabling independent verification of computational methods. The author takes full responsibility for any errors in implementation or interpretation.

Note: This disclosure follows emerging standards for AI transparency in academic research.